

Automatic Metadata Extraction from Scientific Publications Using Artificial Intelligence

Demonstration Publication for AI-Generated Metadata Extraction

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Abstract

This AI-generated demonstration publication illustrates the principles of automatic metadata extraction from scientific papers. It mimics the structure of an academic article and provides a controlled environment for testing metadata recognition, validation, and structuring.

Abstrakti

Tämä tekoälyn tuottama demonstraatiojulkaisu otsikolla: ”Automaattisen metadatan poiminnan haasteet ja mahdollisuudet” esittelee automaattisen metatiedon poiminnan periaatteita tieteellisistä julkaisuista. Se jäljittelee akateemisen artikkelin rakennetta ja tarjoaa testialustan metatiedon tunnistusta, validointia ja rakenteistamista varten.

1 Introduction

Automatic metadata extraction has become a central component of modern digital libraries, research repositories, and large-scale academic indexing systems. As the volume of scientific publications continues to grow, manual metadata curation is no longer feasible. Artificial intelligence offers a scalable and efficient alternative for identifying, structuring, and validating bibliographic information.

This document is an **AI-generated demonstration publication**, designed specifically to test and evaluate automated metadata extraction tools. It does not represent real scientific research but imitates the structure, tone, and conventions of an academic article.

The purpose of this sample is to illustrate how metadata fields can be embedded directly into the document body, enabling software systems to detect and extract them reliably.

2 Background

Metadata extraction traditionally relies on rule-based systems, regular expressions, or manually curated templates. While these approaches work for well-structured

documents, they struggle with heterogeneous formats, multilingual content, and inconsistent layout conventions.

Recent advances in natural language processing have enabled machine learning models to identify metadata fields with significantly higher accuracy.

Transformer-based architectures, in particular, have demonstrated strong performance in recognizing titles, authors, publication years, and identifiers such as DOI or ISBN.

In this demonstration document, the goal is not to present new research findings but to provide a realistic environment for evaluating such systems.

3 Methods

The simulated methodology described here outlines how an AI-based metadata extraction pipeline might operate. Although the details are fictional, they reflect common practices in the field.

The pipeline

This pipeline consists of three conceptual stages:

1. **Document Preprocessing**

The system converts the PDF into machine-readable text, identifies structural elements such as headings, paragraphs, and tables, and normalizes encoding.

2. **Metadata Recognition**

A trained model predicts the location and type of metadata fields. For example, it distinguishes between different identifiers based on contextual cues.

3. **Validation and Structuring**

Extracted metadata is validated against known patterns (e.g., ISSN format) and stored in a structured format such as JSON-LD or Dublin Core.

Throughout this article, metadata fields are intentionally placed in predictable and extractable positions to support testing.

4 Results

Because this is an AI-generated demonstration document, the “results” section does not present empirical findings. Instead, it describes the expected behavior of a metadata extraction system when processing this file.

A well-designed system should:

- correctly identify all metadata fields listed on the metadata page,
- recognize the document structure (title page, sections, references),
- extract identifiers such as DOI, ISSN, and ISBN,
- detect that the document is intentionally labeled as **AI-generated**.

These outcomes help developers verify that their extraction algorithms function as intended.

5 Conclusions

This English-language demonstration publication provides a controlled environment for testing metadata extraction tools. By embedding realistic but fictional metadata fields and structuring the document like an academic article, it enables reliable benchmarking and debugging.

As stated earlier, this is **not a real scientific study**, but an **AI-generated example document** created solely for technical evaluation.

References

Smith, L. (2024). *Neural Approaches to Document Structuring*. Institute of Digital Text Processing.

Kumar, R. & Alvarez, M. (2025). *Automated Indexing in Large-Scale Research Repositories*. Proceedings of the International Conference on Digital Knowledge Systems.

Henderson, P. (2023). *Machine Learning Techniques for Metadata Extraction*. Journal of Computational Information Science, 12(3), 44–59.

Appendix

the prompt used to generate the file:

haluan pdf-tiedoston, joka matkii akateemista julkaisua, mutta jossa mainitaan sen olevan tekoälyn tuotos. haluan, että pdf-tiedostossa on leipätekstiä muutama sivu. haluan, että siinä on otsikkolehti. tiedostolla on tarkoitus demonstroida automaattisesti metatietoja poimivaa ohjelmaa. tiedostossa pitää olla seuraavat metatiedot, jotka voi

ihan vapaasti keksiä. kaikki kentät on oltava tiedostossa itsessään: doctype
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alt_title